Response to Reviewer 2 Comments

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Multi-Granularity Feature Decomposition with Ranking-Aware Regularization for Old Film Restoration

1. Summary

We thank Reviewer 2 for the positive and constructive assessment, and for recognizing that the topic is relevant and timely, the framework technically interesting, the formulation sound, and the results promising. This revised response records: (1) strengthened theoretical justification of representation symmetry; (2) frozen measured ablation/sensitivity evidence centered on F1 (MGFE+RATR, average fusion); (3) frozen complexity evidence using prior 007 round2 L40 unified benchmark artifacts for baseline rows plus a newly verified F1 L40 row; and (4) regenerated RATR visual validation, ROI/video, and failure-case evidence.

2. Questions for General Evaluation

Reviewer's Evaluation	Response and Revisions
Does the introduction provide sufficient background and include all relevant references? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Introduction was shortened and the symmetry-scope argument was strengthened with two formal invariance properties.
Is the research design appropriate? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Yes**	We thank the reviewer. The research design was validated by the F1 main-model selection, the single-protocol retraining, and the extended ablation matrix.
Are the methods adequately described? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. The Methods section now includes the precise definitions of Equations (14) and (23), the insertion-operator implementation, and the reproducibility details.
Are the results clearly presented? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Yes**	We thank the reviewer. The Results section now includes direct RATR validation (branch activation and similarity evolution), the distortion--perception trade-off discussion, and significance testing for LPIPS/DISTS.
Are the conclusions supported by the results? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Yes**	We thank the reviewer. The Conclusions remain conservative and explicitly scoped to the evaluated F1 protocol.
Are all figures and tables clear and well-presented? ( ) Yes ( ) Can be improved ( ) Must be improved ( ) Not applicable Reviewer's rating: **Can be improved**	We thank the reviewer. Figures 4 and 5 were regenerated with larger annotated ROIs; a supplementary video was added.

3. Point-by-point response to Comments and Suggestions for Authors

Comments 1: The main weakness concerns the connection to the journal's scope. The concept of "representation symmetry" is introduced as a balance among restoration branches, but its relationship to traditional notions of symmetry remains unclear. The authors should strengthen the theoretical justification of this concept and clarify how it contributes to symmetry-oriented research.

Response 1:

We thank the reviewer for this core concern and agree that the original wording needed to be more precise. We now adopt a two-layer argument (formal invariance plus honest scope narrowing). Property 1 states that the RATR objective is invariant under branch permutation because DiffKendall is symmetric and the loss sums over unordered branch pairs; Property 2 states that the descriptor average is invariant under forward/backward direction swap. We add a "Symmetry versus balancing" paragraph that distinguishes balancing as an engineered outcome from symmetry as a structural property of the objective. We also add a "Scope of the claim" paragraph honestly stating that we mean functional/representation-level symmetry, not geometric symmetry.

Revision made:

Revised Method Section 3.7 (Property 1/Property 2, symmetry-versus-balancing, scope-of-claim); aligned the Introduction definition of symmetry and updated Related Work Section 2.3.

Comments 2: The validation of the RATR module should be strengthened. The evidence remains largely indirect; additional analyses, such as branch-specific activation visualizations, feature-response studies, or similarity evolution during training, would help demonstrate that the branches learn complementary restoration information rather than merely decorrelated representations.

Response 2:

We thank the reviewer for pushing us to provide more direct evidence. We complement the quantitative ablation with direct visual evidence: branch activation/response maps and training-time branch-similarity evolution curves (A1 MGFE-only versus F1 Full) to show that the branches learn complementary restoration information rather than merely decorrelated representations.

Revision made:

Regenerated Figures 9a, 9b, and 9c from frozen A1/F1 checkpoints using the same 30-clip REDS synthetic protocol. The related Discussion paragraph now reports the direct branch-similarity evidence and cites the figure provenance manifest.

Comments 3: The ablation study deserves deeper discussion. In particular, the MGFE-only configuration improves PSNR while substantially degrading LPIPS and DISTS. This behavior appears consistent with the distortion–perception trade-off and should be discussed in greater detail, particularly to explain why MGFE alone improves reconstruction accuracy while degrading perceptual quality and how RATR mitigates this effect.

Response 3:

We thank the reviewer for pointing to the MGFE-only behavior. After correcting A0 to the same 30-clip protocol, A1 (MGFE-only) improves the baseline but remains weaker than the full F1 model on LPIPS and DISTS. We therefore revised the Discussion to avoid the unsupported "degrades relative to A0" wording and instead explain the evidence-supported conclusion: MGFE recalibration alone is beneficial but insufficient for the best fidelity-perception balance, while RATR plus average fusion gives the strongest combined result.

Revision made:

Expanded the Discussion and updated Table 3 (ablation), now ordered A0/A1/A2/F1(full) with A3 (max fusion) as the fusion-ablation control. Evidence boundary: current numeric table is frozen for measured entries; unmeasured entries are excluded.

Comments 4: Several design choices, including the number of branches, restoration tokens, and the use of progressive max fusion, are only briefly justified. Additional sensitivity analysis would strengthen confidence in the robustness of the proposed framework. Likewise, the claim of modest computational overhead should be supported by quantitative information on training complexity, memory consumption, training time, and parameter growth relative to the baseline model.

Response 4:

We thank the reviewer for requesting stronger sensitivity and complexity justification. Following the D2 model-selection review, the main model is F1 with average fusion, and the earlier max-fusion configuration is retained only as a fusion-ablation control. We report frozen granularity sensitivity across G1 (4t/1b), G2 (8t/2b), F1 (12t/3b, average), and G4 (16t/4b). We also report Table 7 complexity evidence: baseline rows reuse prior 007 round2 L40 unified benchmark artifacts, and the F1 row was newly verified on L40 for this revision with 8.754M parameters, 3107.8G FLOPs per seven-frame clip, 488.5 ms/clip latency, 14.3 FPS, and 0.61 GB peak memory.

Revision made:

sensitivity numbers are frozen for G/R; Table 7 complexity evidence is frozen with row-level provenance.

Comments 5: Reproducibility could be improved by providing additional implementation details, including token initialization, insertion operator implementation, hyperparameter selection, and information regarding code availability.

Response 5:

We thank the reviewer for raising reproducibility. We provide the missing implementation details and a code-availability statement.

Revision made:

Added token initialization strategy (std = 0.02), insertion-operator implementation details, hyperparameter selection rationale, and the planned GitHub release link to the Method section and Appendix.

Comments 6: The Introduction could be shortened by reducing repetitive discussion of representation symmetry and branch balancing. Terminology should be used more consistently throughout the manuscript.

Response 6:

We agree the Introduction contained repetitive symmetry/branch-balancing discussion. We condensed the redundant passages and unified the terminology.

Revision made:

Condensed the Introduction and unified terminology across the manuscript.

Comments 7: The discussion associated with Equations (14) and (23) could be further clarified. The interpretation of the "10% modulation" in Equation (14) would benefit from a more precise mathematical explanation, while Equation (23) relies on ranking information derived from only two temporal neighbors and would benefit from a brief justification of why this signal is sufficient for stable regularization. In addition, statistical significance testing should be extended beyond PSNR and SSIM to include LPIPS and DISTS.

Response 7:

We thank the reviewer for the three detailed points. Equation (14) now gives a precise mathematical definition of the modulation coefficient range and normalization. Equation (23) is justified by arguing that ranking signals from two temporal neighbors are sufficient for stable regularization. We recomputed F1 vs A0 per-clip Wilcoxon tests from A100 logs under the same 30-clip REDS protocol; PSNR, SSIM, LPIPS, and DISTS are all significant, and the Results section now reports effect sizes and per-clip delta summaries.

Revision made:

Clarified Equations (14) and (23) in the Method section; expanded the significance analysis in the Results section.

Comments 8: The enlarged ROI regions in Figures 4 and 5 could be increased in size to facilitate visual comparison of fine restoration details.

Response 8:

We thank the reviewer for the suggestion. We regenerated the qualitative comparison figures with larger, clearly annotated ROIs and added Supplementary Video S1 for temporal real-world comparison.

Revision made:

Updated Figures 4 and 5 with focused high-resolution ROIs. The final figure and video hashes are recorded in `t5-figures/evidence/revision_visual_manifest_20260630.json` and `Paper/symmetry/supplementary/SuppVideoS1_manifest.json`.

Comments 9: The reference list is generally appropriate and relevant; however, a final check for formatting consistency according to MDPI guidelines is recommended.

Response 9:

We thank the reviewer. We performed a final consistency check of the reference list against MDPI guidelines.

Revision made:

Reformatted the reference list in `references.bib` and verified that `latexmk` produces zero bibliography warnings.

Comments 10: The manuscript is written in clear academic English and is generally easy to follow. Nevertheless, moderate proofreading is recommended to improve consistency of terminology, reduce repetition, and shorten several overly long sentences.

Response 10:

We thank the reviewer. After the tables and figures were finalized, we carefully proofread the revised manuscript for terminology consistency, repeated wording, and overly long sentences while preserving all quantitative values.

Revision made:

Proofread the revised manuscript for terminology consistency, concision, and sentence length without changing quantitative values.

4. Response to Comments on the Quality of English Language

Point: The reviewer's assessment of the English language was: The English is fine and does not require any improvement.

We thank the reviewer for this positive assessment. The revised manuscript was nevertheless carefully proofread for terminology consistency, repetition, and sentence length.

5. Additional clarifications

We sincerely thank Reviewer 2 for the constructive and rigorous evaluation, which has substantially improved the manuscript. A marked-up version with all changes highlighted has been uploaded together with the revised manuscript.